

Laws of motion

When a force acts on an object that is free to move, the object will move in accordance with Newton's three laws of motion.

English physicist and mathematician Isaac Newton published his laws of motion in 1687. They explain how objects move—or don't move—and how they react with other objects and forces. These three scientific laws form the basis of what is known as classical mechanics. Modern physics shows that Newton's laws are not perfectly accurate, but they are still useful in everyday situations.

First law of motion

Any object will remain at rest, or move in a straight line at a steady speed, unless an external force acts upon it. So, a soccer ball is stationary until it is kicked and then moves until other forces stop it. This is known as inertia. If all external forces are balanced, the object will maintain a constant velocity. For an object that is not moving, this is zero.



At rest

Gravity acts on the ball, but the ground stops it from moving so it remains at rest.

Force causes motion

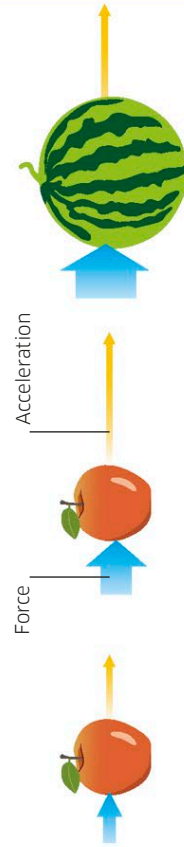
The impact of a cleat kicking the ball applies a force that accelerates the ball.

Force stops motion

The ball slows down due to friction and stops when it meets a cleat.

Second law of motion

When a force acts on an object, the object will generally move in the direction of the force. This causes a change in velocity, known as acceleration. The larger the force, the greater an object's acceleration will be. The more massive an object is, the greater the force needed to accelerate it. This is written as $\text{force} = \text{mass} \times \text{acceleration}$.



Small mass, small force

A force causes an object to accelerate, changing its velocity per second, at a certain rate.

Small mass, double force

If the mass stays the same but the force doubles, the object will accelerate at twice the rate.

Double mass, double force

If the mass doubles and the force doubles again, the rate of acceleration stays the same.

Ariane 5

The Ariane 5 rocket is a launch vehicle used to deliver massive payloads, such as communication satellites, into orbit. Causing a rocket to accelerate upward requires enormous forces to overcome the gravity pulling it downward. Hot gases expand, exerting forces on the walls of the combustion chamber to lift the rocket. The walls of the chamber produce a reaction force that pushes back on the gases, which escape at high speed through the nozzles at the bottom of the engine. These forces create acceleration.

